Task 1: Build a RESTful API (Node.js + Express + MongoDB)

Internship

Alfido Tech – MEAN Stack Developer Internship

Objective

To build a RESTful CRUD API using Node.js, Express.js, and MongoDB with Mongoose for data persistence and validation.

Project Overview

A Task Management API was developed to perform Create, Read, Update, and Delete (CRUD) operations on tasks stored in MongoDB Atlas.

Each task contains:

- Title
- Description
- Status
- Priority
- Created Date
- Updated Date

Technologies Used

- Node.js
- Express.js
- MongoDB Atlas
- Mongoose
- Postman

- Dotenv
- Morgan
- CORS
- Nodemon

Project Structure

alfido-mean-stack/

config/

- db.js

controllers/

- taskController.js

models/

- Task.js

routes/

- taskRoutes.js

.env .env.example server.js

package.json

Installation Commands

npm init -y

npm install express mongoose dotenv cors morgan

npm install –save-dev nodemon

Environment Variables

PORT=3000

MONGO_URI=

MongoDB Connection

The application connects to MongoDB Atlas using Mongoose.

File: config/db.js

Purpose:

- Establish database connection
- Handle connection errors
- Export database connection function

[Screenshot: MongoDB Atlas Cluster]

Task Schema

File: models/Task.js

Fields:

1. title (Required)
2. description (Required)
3. status
    - pending
    - completed
4. priority
    - low
    - medium

- high

Validation was implemented using Mongoose schema validation.

[Screenshot: Task Schema Code]

API Endpoints

1. Create Task

POST /api/tasks

Request Body:

{ “title”: “Build REST API”, “description”: “Complete internship task”, “priority”: “high” }

Response: 201 Created

[Screenshot: POST Request and Response]

---

2. Get All Tasks

GET /api/tasks

Response: 200 OK

[Screenshot: GET All Tasks]

---

3. Get Single Task

GET /api/tasks/:id

Response: 200 OK

[Screenshot: GET Single Task]

---

4. Update Task

PUT /api/tasks/:id

Request Body:

{ “status”: “completed” }

Response: 200 OK

[Screenshot: PUT Request]

---

5. Delete Task

DELETE /api/tasks/:id

Response: 200 OK

[Screenshot: DELETE Request]

Error Handling

Implemented using try-catch blocks.

Handled:

- Invalid requests
- Missing task IDs
- MongoDB connection errors
- Validation errors

[Screenshot: Error Handling Code]

Logging

Morgan middleware was used for request logging.

Example:

GET /api/tasks 200

POST /api/tasks 201

[Screenshot: Terminal Logs]

Results

Successfully implemented a RESTful CRUD API using Node.js, Express.js, and MongoDB.

Features achieved:

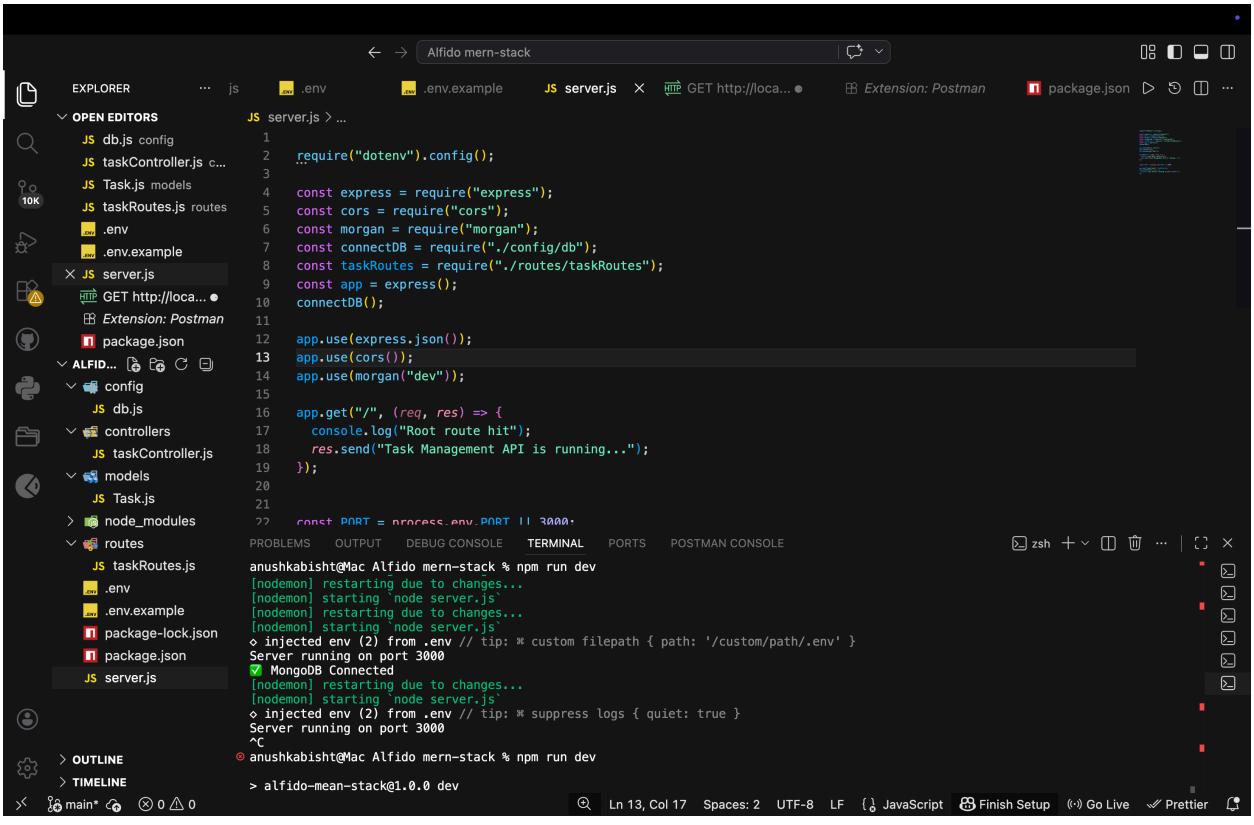
- CRUD Operations
- MongoDB Integration
- Mongoose Validation
- Error Handling
- Logging
- Environment Variables

GitHub Repository

Repository Link:

Conclusion

The project successfully demonstrates backend development concepts including REST API design, MongoDB integration, CRUD operations, validation, logging, and error handling using the MEAN stack ecosystem.


Alfido mern-stack
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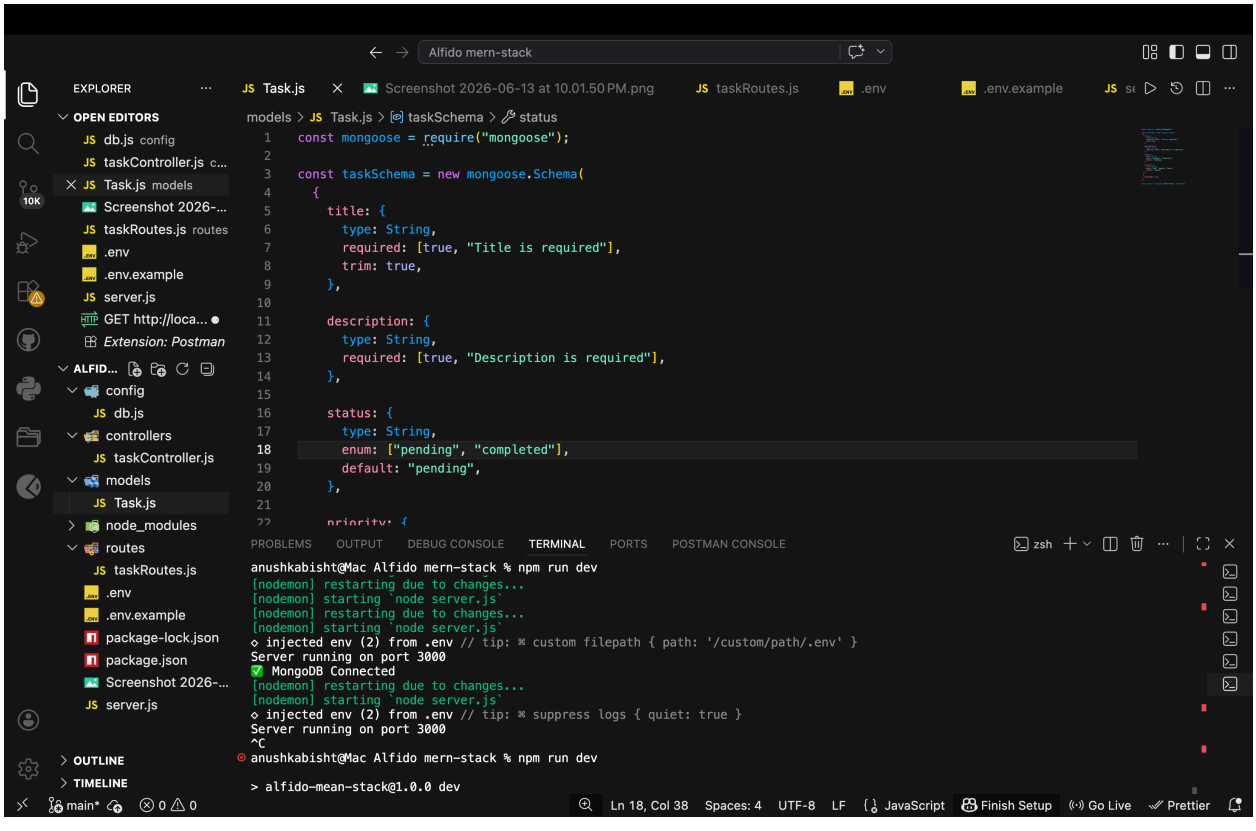
package-lock.json
package.json
server.js
OUTLINE
TIMELINE
server.js
require("dotenv").config();
const express = require("express");
const cors = require("cors");
const morgan = require("morgan");
const connectDB = require("./config/db");
const taskRoutes = require("./routes/taskRoutes");
const app = express();
connectDB();
app.use(express.json());
app.use(cors());
app.use(morgan("dev"));
app.get("/", (req, res) => {
console.log("Root route hit");
res.send("Task Management API is running...");
});
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
POSTMAN CONSOLE
zsh
anushkabisht@Mac Alfido mern-stack % npm run dev
[nodemon] restarting due to changes...
[nodemon] starting `node server.js`
[nodemon] restarting due to changes...
[nodemon] starting `node server.js`
◇ injected env (2) from .env // tip: ⌘ custom filepath { path: '/custom/path/.env' }
Server running on port 3000
✅ MongoDB Connected
[nodemon] restarting due to changes...
[nodemon] starting `node server.js`
◇ injected env (2) from .env // tip: ⌘ suppress logs { quiet: true }
Server running on port 3000
^C
anushkabisht@Mac Alfido mern-stack % npm run dev
> alfido-mean-stack@1.0.0 dev
main*
0 0
Ln 13, Col 17
Spaces: 2
UTF-8
LF
JavaScript
Finish Setup
Go Live
Prettier


Alfido mern-stack
EXPLORER
Task.js
Screenshot 2026-06-13 at 10.01.50 PM.png
taskRoutes.js
.env
.env.example
models > Task.js > taskSchema > status
const mongoose = require("mongoose");
const taskSchema = new mongoose.Schema(
{
title: {
type: String,
required: [true, "Title is required"],
trim: true,
},
description: {
type: String,
required: [true, "Description is required"],
},
status: {
type: String,
enum: ["pending", "completed"],
default: "pending",
},
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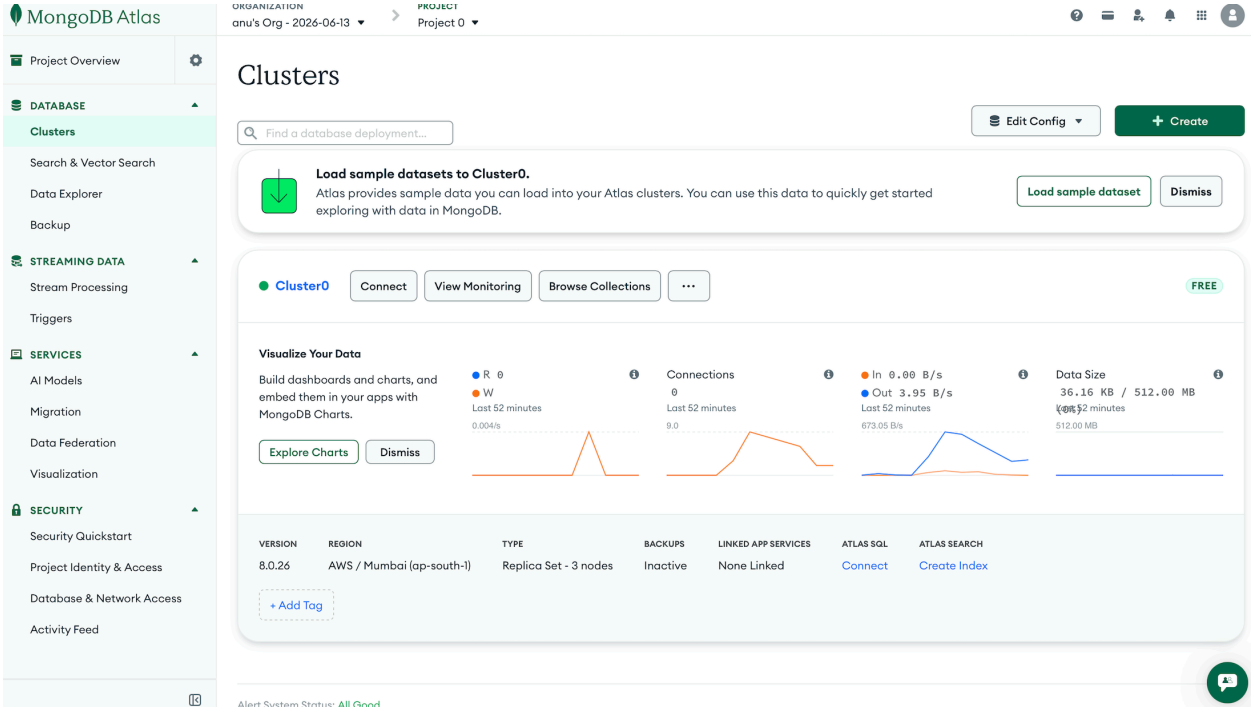
MongoDB Atlas
ORGANIZATION
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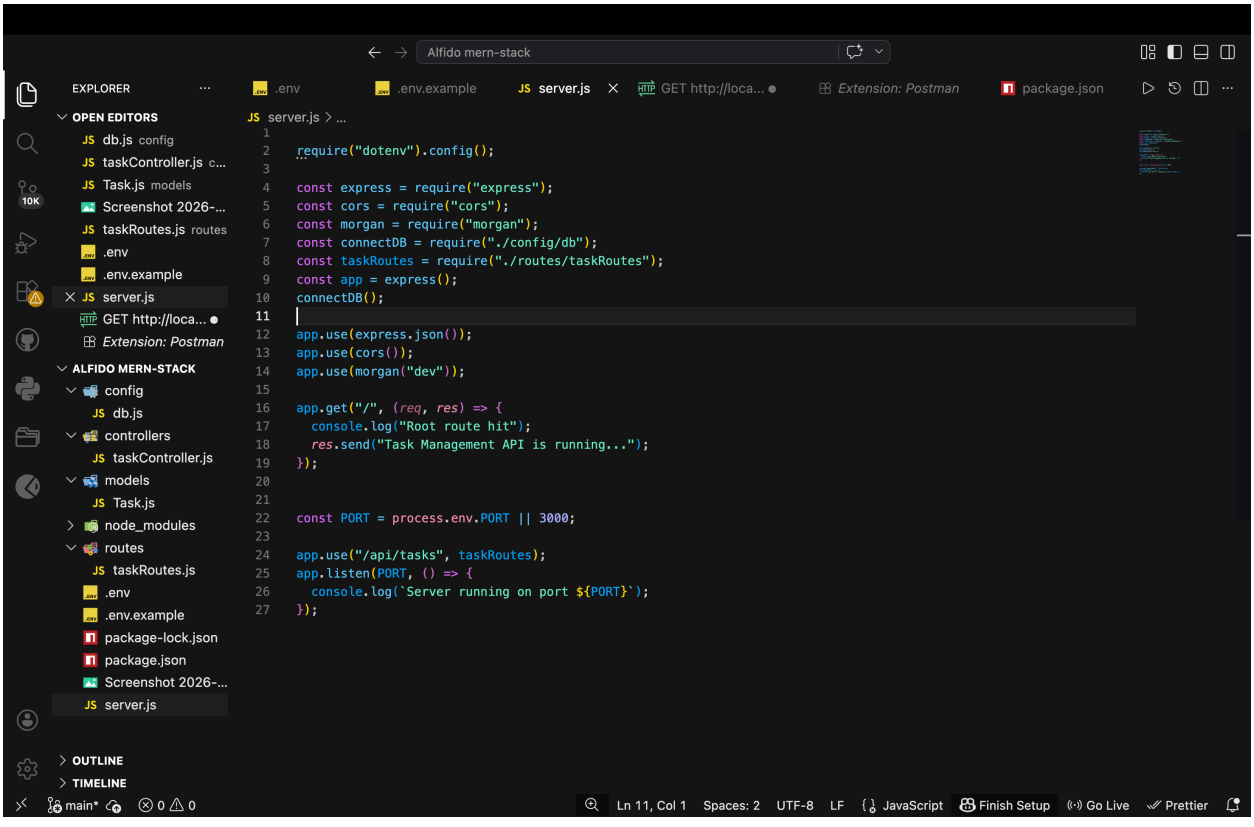
PROJECT
Project 0
Project Overview
DATABASE
Clusters
Search & Vector Search
Data Explorer
Backup
STREAMING DATA
Stream Processing
Triggers
SERVICES
AI Models
Migration
Data Federation
Visualization
SECURITY
Security Quickstart
Project Identity & Access
Database & Network Access
Activity Feed
Clusters
Find a database deployment...
Edit Config
Create
Load sample datasets to Cluster0.
Atlas provides sample data you can load into your Atlas clusters. You can use this data to quickly get started exploring with data in MongoDB.
Load sample dataset
Dismiss
Cluster0
Connect
View Monitoring
Browse Collections
FREE
Visualize Your Data
Build dashboards and charts, and embed them in your apps with MongoDB Charts.
Explore Charts
Dismiss
R 0
W
Last 52 minutes
Connections
0
Last 52 minutes
In 0.00 B/s
Out 3.95 B/s
Last 52 minutes
Data Size
36.16 KB / 512.00 MB
Last 52 minutes
VERSION
8.0.26
REGION
AWS / Mumbai (ap-south-1)
TYPE
Replica Set - 3 nodes
BACKUPS
Inactive
LINKED APP SERVICES
None Linked
ATLAS SQL
Connect
ATLAS SEARCH
Create Index
+ Add Tag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const cors = require("cors");
const morgan = require("morgan");
const connectDB = require("./config/db");
const taskRoutes = require("./routes/taskRoutes");
const app = express();
connectDB();
app.use(express.json());
app.use(cors());
app.use(morgan("dev"));
app.get("/", (req, res) => {
console.log("Root route hit");
res.send("Task Management API is running...");
});
const PORT = process.env.PORT || 3000;
app.use("/api/tasks", taskRoutes);
app.listen(PORT, () => {
console.log(`Server running on port ${PORT}`);
});
main*
0 0
Ln 11, Col 1
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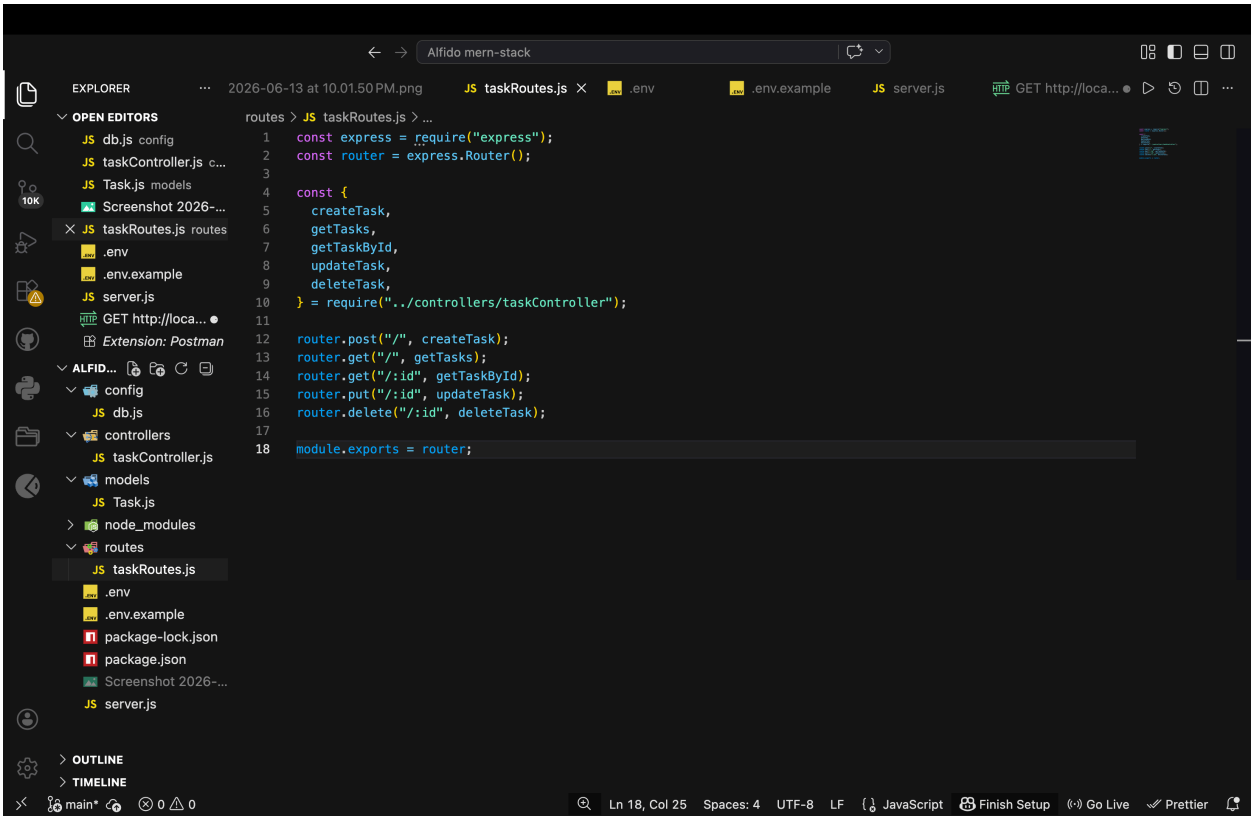
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package-lock.json
package.json
Screenshot 2026-...
server.js
OUTLINE
TIMELINE
routes > taskRoutes.js > ...
const express = require("express");
const router = express.Router();
const {
createTask,
getTasks,
getTaskById,
updateTask,
deleteTask,
} = require("../controllers/taskController");
router.post("/", createTask);
router.get("/", getTasks);
router.get("/:id", getTaskById);
router.put("/:id", updateTask);
router.delete("/:id", deleteTask);
module.exports = router;
main*
0 0
Ln 18, Col 25
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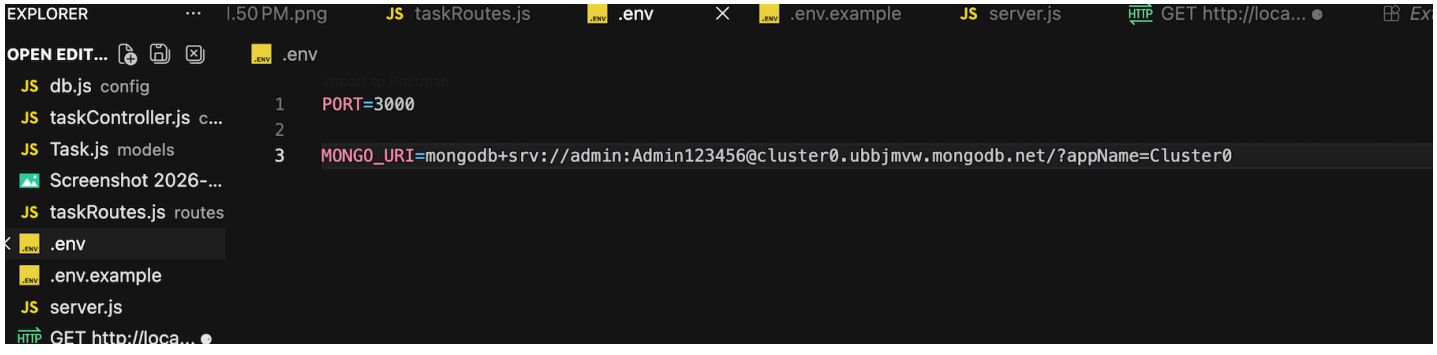
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server.js
GET http://loca...
.env
PORT=3000
MONGO_URI=mongodb+srv://admin:Admin123456@cluster0.ubbjmvw.mongodb.net/?appName=Cluster0